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Department of Electrical Engineering

Class: B.Tech. IV Sem.

Sub: Generation of Electrical Power (EE-224)

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Assignments-2

1. A three-phase 100kW synchronous motor and 3-phase 200 kW induction motor are connected in parallel to a 400 V system. Both motors are star connected. Induction motor is operating at a p.f. of 0.71 lagging. The resistance of synchronous motor is 0.1 ohm per phase and that of the cable, feeding both the motors, is 0.03 ohm per phase. Find power factor of the total load and copper losses in synchronous motor and copper losses in the cable if the power factor of synchronous motor is (a) 1 (b) 0.8 leading (c) 0.6 leading (d) At what power factor are copper losses of synchronous motor minimum (e) At what power factor are total copper losses minimum.
2. A 400 V, 3-phase installation draws a current of 50A at 0.8 lagging power factor. It is desired to install a synchronous motor to improve the overall power factor to 0.95 lagging. The synchronous motor will drive a 25 hp load at an efficiency of 0.9. Find the kVA and power factor of synchronous motor.
3. A factory has an induction motor having a maximum demand of 800 kW at 0.707 pf lagging. The consumer is charged at Rs 80 per kVA of the maximum demand per year. The increase in load is met by installing a synchronous motor of 200 hp and efficiency of 0.99. If the synchronous motor works on full load and at 0.8 pf leading find the difference in annual fixed charges of consumer.
4. A steam plant having an installed capacity of 200 MW is to be set up. The investment on the plant is Rs 9000 per kW of installed capacity. The useful life of plant may be taken as 20 years and salvage value of plant is 25% of initial cost. Find the annul depreciation reserve by (a) straight line method (b) sinking fund method if interest rate is 7%.
5. A consumer has an annual consumption of 176400 kWh. The tariff is Rs 400 per kW of maximum demand per year +60 Paisa per kWh.
 - a) Find his yearly bill and overall cost per kWh if load factor is 36%
 - b) Find overall cost per kWh if the consumption is reduced by 20% with the same load factor
 - c) Find overall cost per kWh if consumption is the same but is 50%.

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